

Extra credit problems

Math 471

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)
1. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.
2. Given distinct points A , B and X make a ruler-and-compass construction of another point $Y \in (AX)$ such that $AY + BY = AX + BX$ and explain why it works. (You may play with the GGeoGebra applet [apfzzybw](#).)